

Linear Simplexification by Square-Quantity Readout and a Dimension-wise Organization of Curvature-Correction Candidates

Subtitle: The positive square map, elementary motion forms, and correction candidates in length, area, and volume

Author: Noriaki Kihara

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Position: First draft as an observational and organizing paper. The possible originality does not lie in the square map itself, but in the way the square-quantity readout can be placed side by side with the dimension-wise results of the self-referenced Paper 0, thereby organizing where curvature-correction candidates first become relevant among length, area, and volume. This paper does not claim to prove physical laws, assert observational facts, or replace established physics.

Self-reference: Noriaki Kihara, “Paper 0: Distortion of the Geodesic Unit Cell in Positively-Curved Constant-Curvature Space”, v1.4.

Abstract

This paper formulates, using only elementary geometry and elementary algebra, the following fact. If the square quantities

$$X_i = x_i^2$$

are read as the basic variables on the positive real domain, then the curved sum-of-squares constraint

$$\sum_i x_i^2 = E$$

is read, in square-quantity space, as the linear constraint

$$\sum_i X_i = E.$$

This fact itself has known relatives, for example the square-root representation connecting a probability simplex with the positive orthant of a sphere in information geometry. Therefore, this paper does not claim the correspondence between the simplex and the positive spherical orthant as a new theorem.

The aim of this paper is to place this square-root correspondence, which is close to known constructions, not as a geometry of probability densities but as a general “square-quantity readout”, and then to observe at which geometric dimension a curvature-correction candidate first becomes relevant. The square-quantity readout does

not make curvature disappear. Rather, one sees a branching: no correction candidate appears in relations closed by length alone, while the area-correction coefficient

$$k_s(R)$$

from the self-referenced Paper 0 appears as a test candidate when a two-dimensional area quantity first enters.

More concretely, first we confirm that the positive square map sends sum-of-squares constraints such as a circular arc or a spherical octant exactly to linear constraints such as a line segment or a triangular simplex. Second, we place simple equations on the square-quantity side and see that, under positive square-root readout, equations isomorphic to uniform-motion and uniformly-accelerated-motion forms are obtained. Third, even if masses and conservation laws are externally imposed, as in a two-body collision, no area correction appears as long as the situation is closed in one dimension. Fourth, a centrifugal-type model, in which tangential and radial directions appear simultaneously, is treated as the first natural test example in which an area quantity appears.

This paper states the above as observations and elementary statements in geometry and algebra. Equations from classical physics are used only as examples for asking: if a known form is placed on the square-quantity side or on the square-root side, which geometric dimension of correction becomes relevant? They are not used as derivations of physical laws.

0. What This Paper Claims and Does Not Claim

0.1 Claims

The claims of this paper are limited to the following.

1. On the positive domain, the square map

$$\Phi : (x_1, \dots, x_n) \mapsto (X_1, \dots, X_n) = (x_1^2, \dots, x_n^2)$$

maps a sum-of-squares constraint to a linear simplex constraint.

2. Its inverse is the positive square-root map

$$\Psi : (X_1, \dots, X_n) \mapsto (\sqrt{X_1}, \dots, \sqrt{X_n}),$$

and the correspondence is one-to-one in the interior domains $x_i > 0$, $X_i > 0$.

3. If simple equations such as

$$X = AT, \quad X = \frac{1}{2}BT^2$$

are placed on the square-quantity side, then positive square-root readout gives

$$x = \sqrt{A}t, \quad x = \sqrt{B/2}t^2.$$

4. According to the self-referenced Paper 0, a geodesic edge length in a constant-curvature space is not distorted by the construction, while two-dimensional area is distorted. Hence a branching rule arises: no area correction is inserted into a one-dimensional relation closed by length alone, and $k_s(R)$ first becomes a candidate when an area cell spanned by two directions appears.
5. A centrifugal-type model is treated as the first natural test example in which an area quantity appears. The insertion of $k_s(R)$ proposed there is not a physical law uniquely derived from geometry, but a model readout under the assumptions of constant curvature and uniform correction.

0.2 Non-claims

This paper does not claim the following.

- It does not claim to prove physical laws.
- It does not claim to replace classical mechanics, relativity, or quantum theory.
- It does not identify X_i with a particular physical quantity such as probability, energy, mass, or time.
- It does not claim that the square map is an isometry as a metric map.
- It does not claim that a curved space itself disappears into a flat space.
- It does not claim that inserting $k_s(R)$ into a centrifugal-type equation is uniquely derived as a law of mechanics.
- It does not claim that the non-isometry of the square map is the cause of $k_s(R)$.
- It does not identify the metric distortion of the square map with the geodesic-square area excess of the self-referenced Paper 0.
- It does not reject Riemannian geometry or local differential geometry.

When this paper says that a relation is “read as flat”, it means only that the constraint

$$\sum_i x_i^2 = E$$

becomes, on the square-quantity side, the affine linear equation

$$\sum_i X_i = E.$$

This is not an identity of metrics, but an algebraic simplification in the readout coordinates.

0.3 Precise Position of This Paper

This paper is not an alternative theory to Riemannian geometry for general continuous systems with local curvature. It restricts the setting to constant-curvature spaces with a uniform curvature radius, and to cases where geodesic unit cells can be considered.

Under this restriction, the question is what can be read using only elementary geometry and elementary algebra. The self-referenced Paper 0 studied how edge length, vertex angle, area, and volume change with respect to the curvature radius R . As a sequel, the present paper asks, within the square-quantity readout, which kinds of equations are closed by length alone and which kinds require a two-dimensional area cell.

Thus the central observation is not that curvature disappears. The central observation is the dimension-wise organization that no area-correction candidate appears in relations closed by length alone, while a correction

candidate first appears when an area quantity is involved. This is a juxtaposition of several independent geometric facts, not a claim that $k_s(R)$ is causally derived from the non-isometry of the square map.

1. The Positive Square Map

1.1 The Two-Dimensional Case

First consider two dimensions. In the positive quadrant

$$x \geq 0, \quad y \geq 0,$$

take the unit circular arc

$$x^2 + y^2 = 1.$$

Set

$$X = x^2, \quad Y = y^2.$$

Then immediately

$$X + Y = 1.$$

Thus a point on the positive circular arc is read, on the square-quantity side, as a point on a straight line simplex.

Fig. 1. Squaring sends a curved positive arc to a linear simplex edge

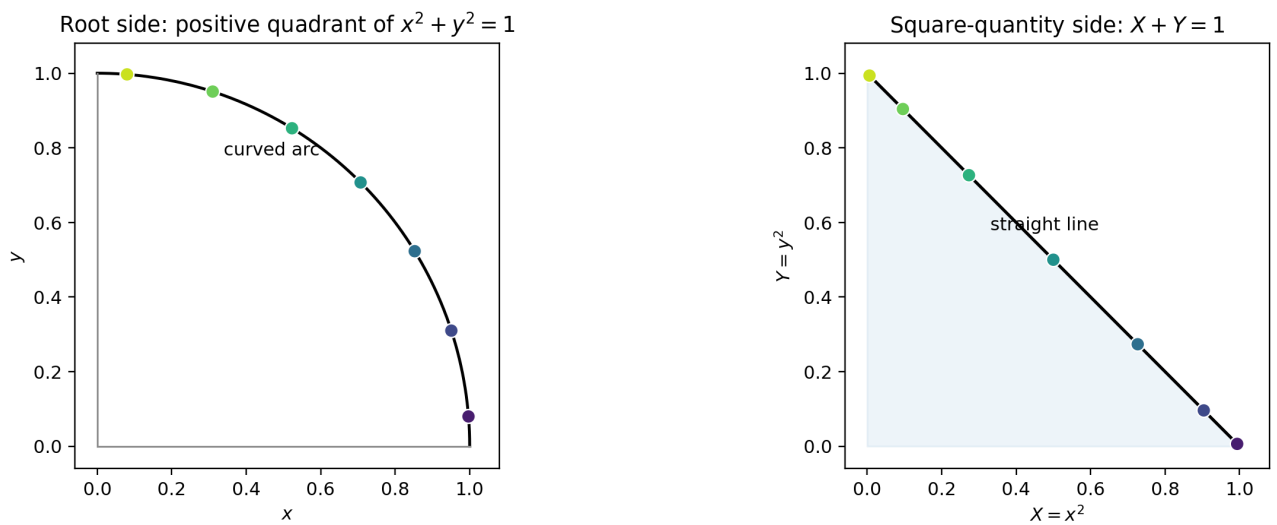


Fig. 1

Figure 1. The positive circular arc $x^2 + y^2 = 1$ is read on the square-quantity side as the line simplex $X + Y = 1$.

This is not an approximation. For example, if

$$(x, y) = \left(\frac{3}{5}, \frac{4}{5} \right),$$

then

$$(X, Y) = \left(\frac{9}{25}, \frac{16}{25} \right),$$

and indeed

$$\frac{9}{25} + \frac{16}{25} = 1.$$

Conversely, if

$$(X, Y) = \left(\frac{9}{25}, \frac{16}{25} \right)$$

is given on the square-quantity side, taking positive square roots gives

$$(x, y) = \left(\frac{3}{5}, \frac{4}{5} \right).$$

The restriction to the positive domain is important. If signs are allowed, the same value $X = x^2$ returns both x and $-x$. In this paper, in order to make the square-root readout unique, we basically work on the positive domain.

1.2 General Dimension

The same statement holds in arbitrary dimension.

On the positive domain

$$x_i \geq 0 \quad (i = 1, \dots, n),$$

consider the sum-of-squares constraint

$$S_E^{n-1,+} = \left\{ (x_1, \dots, x_n) \mid \sum_{i=1}^n x_i^2 = E, \ x_i \geq 0 \right\}.$$

On the square-quantity side, set

$$X_i = x_i^2$$

and consider

$$P_E^{n-1} = \left\{ (X_1, \dots, X_n) \mid \sum_{i=1}^n X_i = E, X_i \geq 0 \right\}.$$

Then

$$\Phi : S_E^{n-1,+} \rightarrow P_E^{n-1}, \quad \Phi(x_1, \dots, x_n) = (x_1^2, \dots, x_n^2)$$

is a one-to-one correspondence, and its inverse is

$$\Psi : P_E^{n-1} \rightarrow S_E^{n-1,+}, \quad \Psi(X_1, \dots, X_n) = (\sqrt{X_1}, \dots, \sqrt{X_n}).$$

Proof.

If $(x_1, \dots, x_n) \in S_E^{n-1,+}$, then each $X_i = x_i^2$ is nonnegative and

$$\sum_i X_i = \sum_i x_i^2 = E,$$

so $\Phi(x) \in P_E^{n-1}$.

Conversely, if $(X_1, \dots, X_n) \in P_E^{n-1}$, then each $X_i \geq 0$, so $\sqrt{X_i}$ is defined, and

$$\sum_i (\sqrt{X_i})^2 = \sum_i X_i = E.$$

Therefore $\Psi(X) \in S_E^{n-1,+}$.

Furthermore, on the positive domain,

$$\Psi(\Phi(x)) = (\sqrt{x_1^2}, \dots, \sqrt{x_n^2}) = (x_1, \dots, x_n),$$

and

$$\Phi(\Psi(X)) = ((\sqrt{X_1})^2, \dots, (\sqrt{X_n})^2) = (X_1, \dots, X_n).$$

Hence the two maps are inverse to each other. $\square$

1.3 The Three-Dimensional Case

In three dimensions, the positive spherical octant

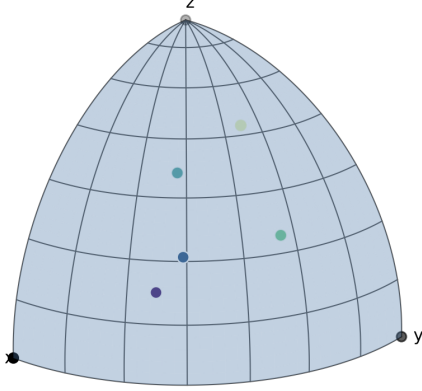
$$x^2 + y^2 + z^2 = 1, \quad x, y, z \geq 0$$

is read on the square-quantity side as the standard two-simplex

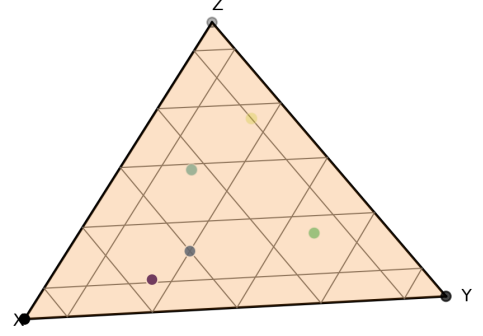
$$X + Y + Z = 1, \quad X, Y, Z \geq 0.$$

Fig. 2. In 3D, the positive spherical octant reads as the standard simplex

Root side: octant of $x^2 + y^2 + z^2 = 1$



Square-quantity side: $X + Y + Z = 1$



The same map $(X, Y, Z) = (x^2, y^2, z^2)$ changes a spherical constraint into an affine plane; matched colors mark exact corresponding points.

图 2 Fig. 2

Figure 2. The positive spherical octant is mapped to the standard simplex on the square-quantity side. Points of the same color represent the same point read through the square map.

Again, the important point is that a curved constraint on the spherical side becomes a triangle on a plane on the square-quantity side. The figure is not schematic; it is generated from the map

$$(X, Y, Z) = (x^2, y^2, z^2).$$

1.4 This Is Not an Isometric Flattening

Here one must be careful. The square map is generally not an isometry.

Taking differentials gives

$$dX_i = 2x_i dx_i,$$

so lengths are not preserved as they are. Therefore the intrinsic metric of the sphere does not simply become the Euclidean metric of the plane.

The fact used in this paper is more limited.

The sum-of-squares constraint

$$\sum_i x_i^2 = E$$

becomes, on the square-quantity side,

$$\sum_i X_i = E.$$

This single algebraic point is the basis of the readouts below.

2. Placing Equations on the Square-Quantity Side

2.1 Basic Setting

We now extract two square quantities

$$X = x^2, \quad T = t^2.$$

Here X and T are variables on the square-quantity side, while x and t are their positive square-root-side variables.

This paper does not immediately identify T with physical time. It is simply one coordinate on the square-quantity side. However, because the resulting readout has the same form as familiar classical motion equations, the terms “uniform-motion form” and “uniformly-accelerated-motion form” are used as comparisons.

2.2 A Linear Equation Gives the Uniform-Motion Form

Place

$$X = AT$$

on the square-quantity side, where $A > 0$ is constant.

To return to the square-root side, substitute

$$X = x^2, \quad T = t^2.$$

Then

$$x^2 = At^2.$$

Taking the positive square root gives

$$x = \sqrt{A}t.$$

This has the same form as ordinary uniform motion

$$x = vt.$$

However, this paper does not call it a derivation of a physical velocity v . It only states the algebraic fact that one may read

$$v_{\text{read}} = \sqrt{A}.$$

2.3 Double-Cone-Type Readout

The equation in the previous subsection may also be written as

$$x^2 - At^2 = 0.$$

If $A = c^2$, then

$$x^2 - c^2t^2 = 0.$$

Geometrically, this is a double-cone-type quadratic form, and in a two-dimensional section it is a pair of straight lines. This paper treats this equation not as a physical metric condition, but as a geometric form that appears when a linear equation on the square-quantity side is read on the square-root side.

2.4 A Quadratic Equation Gives the Uniformly-Accelerated-Motion Form

Next place

$$X = \frac{1}{2}BT^2$$

on the square-quantity side, where $B > 0$ is constant.

Again substituting

$$X = x^2, \quad T = t^2$$

gives

$$x^2 = \frac{1}{2}Bt^4.$$

Taking the positive square root gives

$$x = \sqrt{\frac{B}{2}} t^2.$$

This has the same form as ordinary uniformly accelerated motion

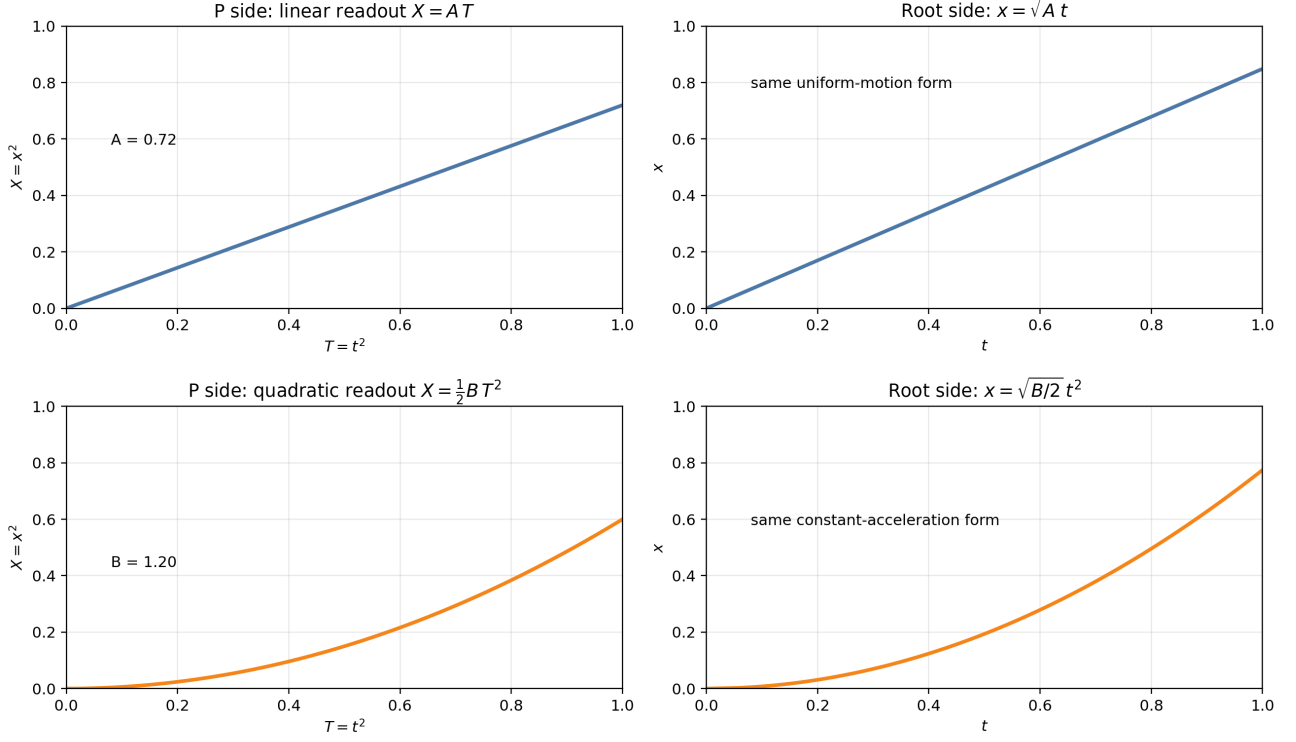
$$x = \frac{1}{2}at^2.$$

If one compares the coefficients, then

$$a_{\text{read}} = \sqrt{2B}.$$

Again, no physical acceleration is being proved. The statement is only the algebraic observation that a quadratic equation on the square-quantity side becomes the uniformly-accelerated-motion form under positive square-root readout.

Fig. 3. Elementary motion forms are recovered by square-root readout



No physical law is assumed here; these are algebraic readouts of equations placed on the square-quantity side.

图3 Fig. 3

Figure 3. A linear equation and a quadratic equation placed on the square-quantity side become, under positive square-root readout, uniform-motion and uniformly-accelerated-motion forms.

2.5 General Form

More generally, place

$$X = CT^m$$

on the square-quantity side, where $C > 0$ and $m > 0$.

Since

$$X = x^2, \quad T = t^2,$$

we have

$$x^2 = Ct^{2m}.$$

Taking the positive square root gives

$$x = \sqrt{C} t^m.$$

Thus the power m on the square-quantity side appears as the same power m on the square-root side. Only the coefficient is square-rooted.

This is a small-looking fact, but it is important. Although X and T are both square quantities on the square-quantity side, the familiar power relation itself remains unchanged on the positive square-root side.

3. Why Curvature Correction Does Not Appear in One-Dimensional Equations

The self-referenced Paper 0 established the following distinction for geodesic unit cells placed in a constant-curvature space.

- A geodesic edge length is exactly preserved by the construction.
- Vertex angle and two-dimensional area change with the curvature radius R .
- Volume changes in a dimension-dependent way.

Moreover, Paper 0 showed that, in the small-curvature limit, the excess coefficient of d -dimensional volume is

$$V_d(R) = 1 + \frac{c_d}{R^2} + O(R^{-4}), \quad c_d = \frac{d(d-1)}{12} = \frac{\binom{d}{2}}{6}.$$

This means that the curvature excess of d -dimensional volume can be read as the sum of the area excesses of the $\binom{d}{2}$ coordinate two-planes.

From this viewpoint, the first carrier of curvature is not length but area. In a one-dimensional relation closed by length alone, no area correction enters.

The equations

$$x = \sqrt{A} t$$

and

$$x = \sqrt{B/2} t^2$$

from the previous section are both closed, after readout, as relations between one x and one t . No two-dimensional area cell appears.

Therefore, in the reading of this paper, there is no immediate reason to multiply either the uniform-motion form or the uniformly-accelerated-motion form by $k_s(R)$.

The observation obtained here is:

Whether a curvature correction appears depends not on whether “acceleration” is present, but on whether a two-dimensional area quantity is present.

This sentence should be read not as a law of mechanics, but as a geometric-dimensional test rule. There is no reason to mechanically attach the area coefficient of a two-dimensional geodesic cell to an equation closed by one-dimensional quantities alone. Conversely, when two independent directions appear simultaneously and the equation implicitly contains a two-dimensional cell, the possibility of considering an area coefficient first arises.

4. What Happens in a Two-Body Collision Equation

The framework of this paper itself does not contain mass. Therefore, rather than deriving mass from the framework, we only ask what happens if masses are externally assigned.

On a one-dimensional line, assign two masses

$$m_1, \quad m_2$$

to two objects. Let the pre-collision velocities be

$$u_1, \quad u_2$$

and the post-collision velocities be

$$u'_1, \quad u'_2.$$

For an ordinary one-dimensional perfectly elastic collision, one places

$$m_1 u_1 + m_2 u_2 = m_1 u'_1 + m_2 u'_2$$

and

$$\frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2 = \frac{1}{2} m_1 u'^2_1 + \frac{1}{2} m_2 u'^2_2.$$

Solving these gives

$$u'_1 = \frac{m_1 - m_2}{m_1 + m_2} u_1 + \frac{2m_2}{m_1 + m_2} u_2,$$

$$u'_2 = \frac{2m_1}{m_1 + m_2} u_1 + \frac{m_2 - m_1}{m_1 + m_2} u_2.$$

Here again, $k_s(R)$ does not appear.

However, the energy equation contains u^2 . Is this not an area quantity?

The answer of this paper is limited as follows.

Here u^2 is the square of the same one-dimensional velocity component, not an area cell spanned by two independent directions. The whole equation is closed by velocity variables on a one-dimensional line. Therefore there is no place to insert the “distortion of two-dimensional area” from the self-referenced Paper 0.

This point is important for the organizing principle of the paper. Even when a two-body problem, mass, momentum, and energy appear, curvature correction does not appear for that reason alone. The entrance to correction is not the number of objects, but the appearance of an area quantity.

5. A Centrifugal-Type Model in Which an Area Quantity Appears

5.1 The Centrifugal-Type Equation

Next consider a centrifugal-type equation.

Classically, for radius r and speed v , the magnitude of centrifugal acceleration is written as

$$a_c = \frac{v^2}{r}.$$

If a mass m is externally assigned, then

$$F_c = m \frac{v^2}{r}.$$

This paper does not derive this as a physical law. It uses the known classical form only as an example for asking where an area correction should be considered when the form is placed within the square-quantity readout.

5.2 Why Area Appears Here

The uniform-motion form

$$x = vt$$

and the uniformly-accelerated-motion form

$$x = \frac{1}{2}at^2$$

were, in the present readout, closed as relations along one line.

In a centrifugal-type equation, however, the tangential speed v and the radial length r appear simultaneously. Geometrically, circular motion involves two directions: the tangential direction and the radial direction. Here, for the first time, information of a two-dimensional cell, rather than length alone, is required.

For this reason, the centrifugal-type equation is the first natural test example in this paper where an area-correction coefficient appears. What follows from this is only that there is an entrance for considering an area coefficient. Which dynamical quantity it multiplies, in which direction, and to which power are separate questions.

5.3 The Area Coefficient from the Self-Referenced Paper 0

In the self-referenced Paper 0, for a geodesic square of side length 1 in a positively curved constant-curvature space with curvature radius R , the vertex angle

$$\theta(R) = \arccos\left(-\tan^2 \frac{1}{2R}\right)$$

was obtained.

The area of this geodesic square is

$$A(R) = R^2\{4\theta(R) - 2\pi\}.$$

Since the area of the flat unit square is 1, this paper defines the area-correction coefficient as

$$k_s(R) := A(R),$$

namely

$$k_s(R) = R^2 \left[4 \arccos\left(-\tan^2 \frac{1}{2R}\right) - 2\pi \right].$$

For a positively curved geodesic square of side length 1, at least

$$R \geq \frac{2}{\pi}$$

is required for existence. In the flat limit,

$$\lim_{R \rightarrow \infty} k_s(R) = 1,$$

and the leading expansion is

$$k_s(R) = 1 + \frac{1}{6R^2} + O(R^{-4}).$$

Figure 5. The area-correction coefficient $k_s(R)$ of a geodesic square of side length 1, obtained from the formula of the self-referenced Paper 0.

5.4 Test Insertion into the Centrifugal-Type Model

From the above, one test model readout of the centrifugal-type equation containing area correction is

$$a_c^{(R)} = k_s(R) \frac{v^2}{r}.$$

If a mass m is externally assigned, then

Fig. 5. Exact area coefficient for a unit geodesic square

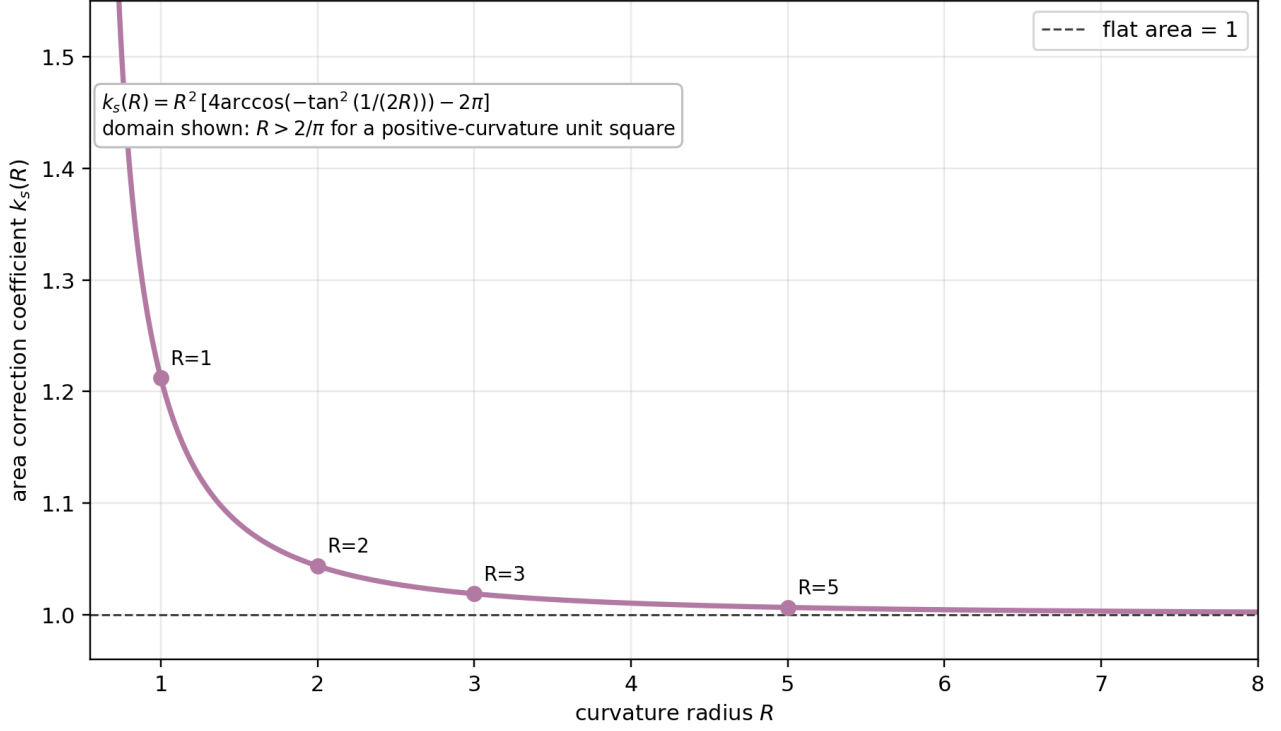


Fig. 5

$$F_c^{(R)} = m k_s(R) \frac{v^2}{r}.$$

Here $k_s(R)$ is not an adjustable free parameter. Once the curvature radius R is fixed, it is uniquely determined by the geometric formula of the self-referenced Paper 0.

However, the fact that $k_s(R)$ is uniquely determined is different from the claim that it must be inserted into a centrifugal-type equation in precisely the direction written above. Depending on whether one reads the quantity as a measure, as a density, as an area, or as an inverse area, $k_s(R)$, $1/k_s(R)$, or another function may appear.

Thus the meaning of this equation is limited. It is not a physical proof of centrifugal force, but a minimal model in which the classical centrifugal-type equation is placed in a geometric readout with area correction. What this paper fixes is only that the centrifugal-type form contains two directions and, unlike one-dimensional equations, has a natural entrance for considering area correction.

6. Geometric Vocabulary: Elliptic, Double-Cone, and Hyperbolic Types

Quadratic forms appearing under square-quantity readout fall into several geometric types depending on the signs and the right-hand side.

$$x^2 + y^2 = 1$$

is elliptic, and in two dimensions it is a circle.

$$x^2 - y^2 = 0$$

is double-cone type, and in a two-dimensional section it is a pair of lines.

$$x^2 - y^2 = 1$$

is hyperbolic.

Fig. 4. Different root-side geometries can share a simple quadratic-form origin

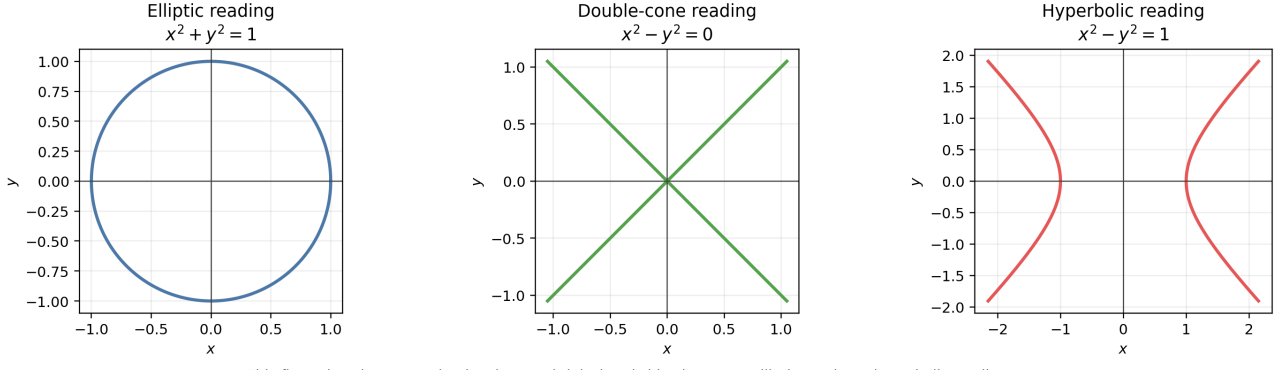


图 5 Fig. 4

Figure 4. Elliptic, double-cone, and hyperbolic readouts appear from sign choices in the same family of quadratic forms.

This paper uses these not as types of physical space, but as vocabulary for pure quadratic curves and surfaces appearing on the square-root side.

7. Relation to Known Structures and the Scope of This Paper's Originality

7.1 Known Parts

It is known, in a broad sense, that a positive square-root or square map relates a simplex to the positive orthant of a sphere.

For example, in information geometry, if one takes

$$q_i = \sqrt{p_i}$$

for probability densities p_i , then

$$\sum_i p_i = 1$$

becomes

$$\sum_i q_i^2 = 1.$$

In this square-root density representation, a probability simplex is treated as a positive spherical region, and the relation between the Fisher–Rao metric and spherical geometry is studied.

Therefore, this paper does not claim the square-root correspondence between a simplex and a positive spherical orthant itself as a new discovery.

7.2 The Organizing Principle of This Paper

The possible originality of this paper does not lie in the square map itself. The square map is an entrance and is close to known structures. The possible contribution of this paper is the organizing principle that asks, after the square-quantity readout is adopted, “where should a curvature-correction candidate be considered?” in a dimension-wise manner.

More concretely, the contribution is the following combination.

1. Instead of probability densities, general square quantities $X_i = x_i^2$ are placed as the basic readout.
2. The fact that a curved sum-of-squares constraint becomes a linear simplex constraint on the square-quantity side is formulated as algebra and geometry without physical interpretation.
3. A linear and a quadratic equation placed on the square-quantity side are shown to become the uniform-motion and uniformly-accelerated-motion forms under positive square-root readout.
4. Even when masses and conservation laws are placed, as in a two-body collision, no area correction appears as long as the relation is closed in one dimension.
5. Using the structure from the self-referenced Paper 0 that “the seat of distortion lies at $d \geq 2$ ”, relations involving only length are separated from relations involving an area cell.
6. A centrifugal-type model, in which an area quantity first appears, is positioned as the first natural test example for the appearance of $k_s(R)$. However, the way of multiplying by $k_s(R)$ is not claimed to have been uniquely derived as a law of mechanics.

If an existing paper were found that states this entire combination in the same form, namely square-quantity readout, linear simplexification, absence of correction in one-dimensional equations, appearance of a correction candidate in two-dimensional area cells, and connection to the geodesic-square coefficient $k_s(R)$ in constant curvature, then the present paper should be positioned not as a new organizing principle but as a rediscovery. In the range checked so far, square-root density representations and Fisher–Rao geometry are close, but no source has been confirmed that states this dimension-wise organization of curvature-correction candidates in the same form.

8. Conclusion

This paper formulated the positive square map

$$X_i = x_i^2$$

as a basic readout, and showed that the sum-of-squares constraint

$$\sum_i x_i^2 = E$$

becomes the linear simplex constraint

$$\sum_i X_i = E$$

on the square-quantity side.

From this simple correspondence, the linear equation

$$X = AT$$

placed on the square-quantity side becomes, on the positive square-root side,

$$x = \sqrt{A}t,$$

and the quadratic equation

$$X = \frac{1}{2}BT^2$$

becomes

$$x = \sqrt{B/2}t^2.$$

These are isomorphic to the classical uniform-motion and uniformly-accelerated-motion forms, but this paper does not call them derivations of physical laws.

Also, even if the equations for a two-body perfectly elastic collision are externally placed, no area correction appears as long as the equations are closed by one-dimensional velocities alone. This indicates that the entrance to curvature correction is not “acceleration” or “mass” itself, but a two-dimensional area quantity.

Finally, in a centrifugal-type model, tangential and radial directions appear simultaneously, and a natural entrance for considering a two-dimensional area cell arises. In that case, the exact formula from the self-referenced Paper 0 gives

$$k_s(R) = R^2 \left[4 \arccos \left(-\tan^2 \frac{1}{2R} \right) - 2\pi \right].$$

This coefficient is not an arbitrary adjustment coefficient, but is defined as the area of a geodesic square of side length 1. However, how it should be reflected in a centrifugal-type mechanical equation is a separate issue. This paper only places, as a minimal test model,

$$a_c^{(R)} = k_s(R) \frac{v^2}{r}.$$

This is the scope of the paper. Square-quantity readout juxtaposes a curved sum-of-squares constraint, a linear simplex constraint, elementary motion forms, and an area-correction coefficient within one elementary algebraic-geometric flow. Its core is neither to erase curvature nor to derive area correction from the non-isometry of the square map. It is to organize, dimension by dimension, the difference between relations closed by length alone and relations for which a two-dimensional area cell should be considered.

References

Self-reference

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